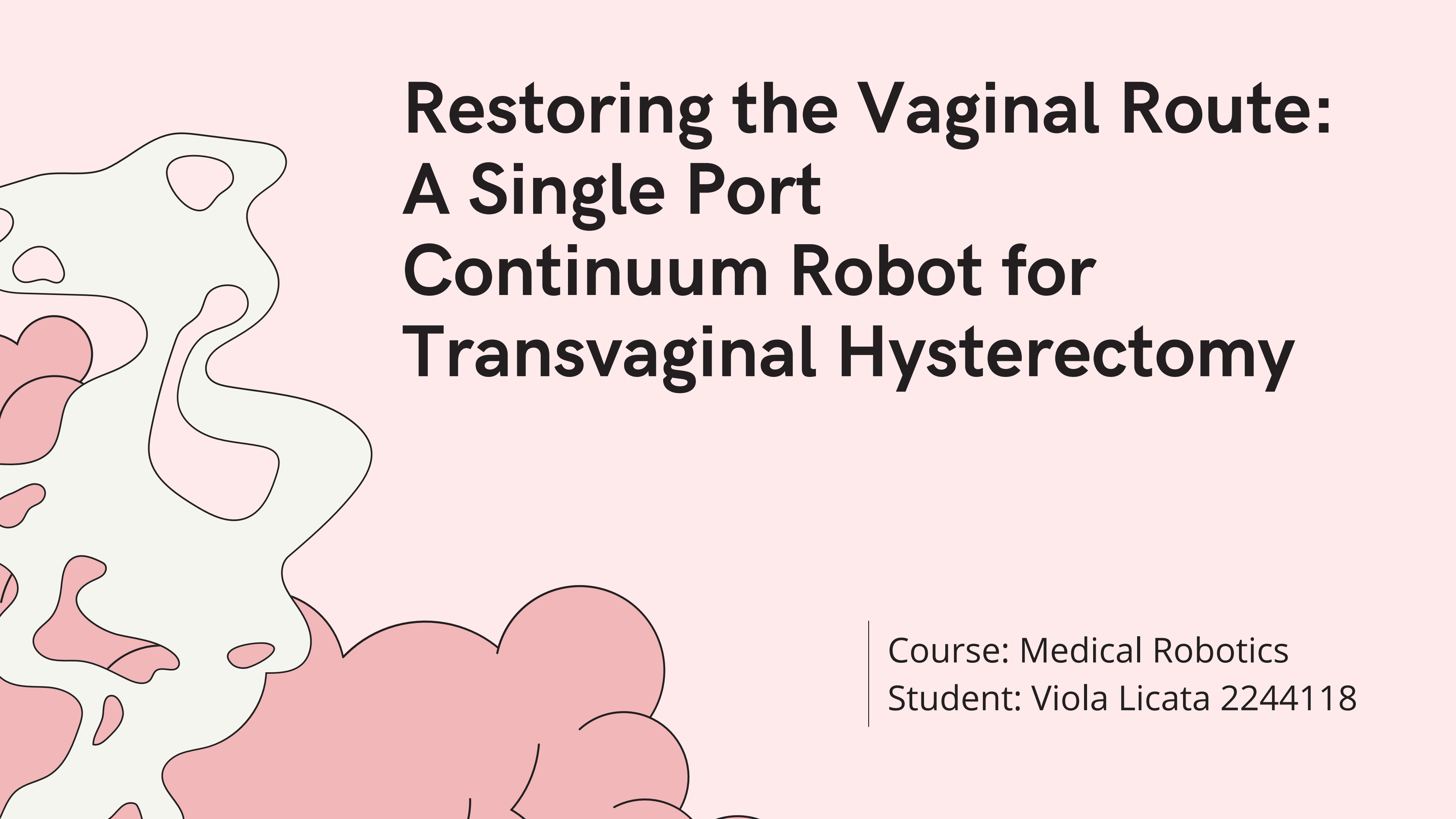




Restoring the Vaginal Route: A Single Port Continuum Robot for Transvaginal Hysterectomy

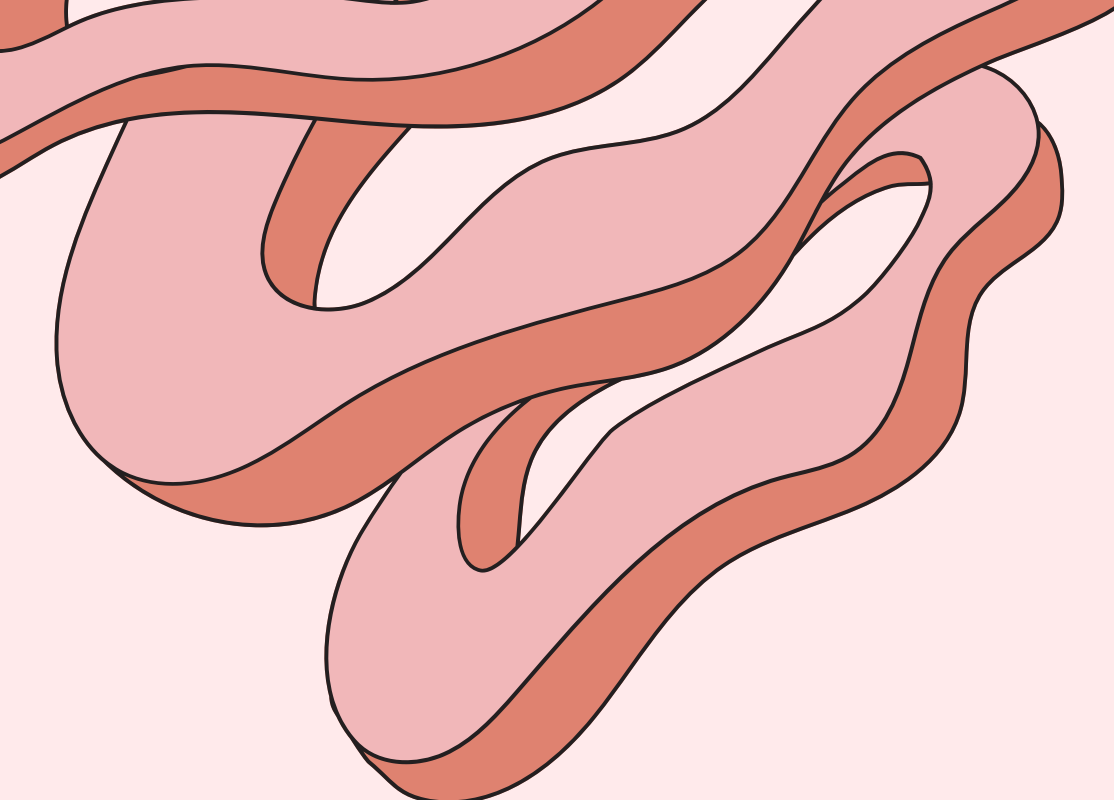
Course: Medical Robotics
Student: Viola Licata 2244118

The Problem:

Over 800,000 women each year undergo **invasive** abdominal surgery for procedures that could be performed through a natural orifice.

The Need:

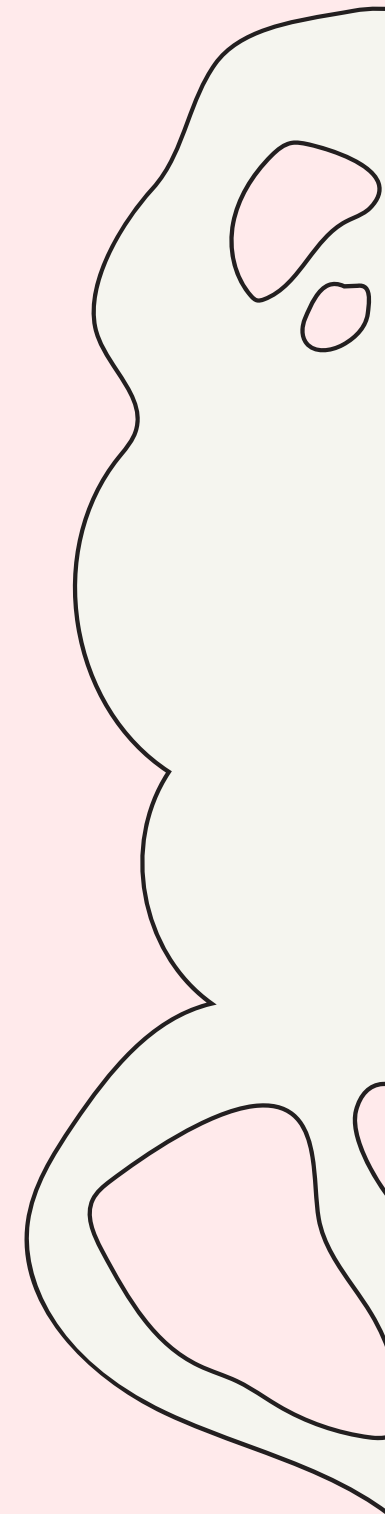
The lack of suitable instruments forces surgeons into abdominal approaches, increasing patient risk, surgical trauma, and healthcare costs. The goal is to make natural-orifice hysterectomy feasible.

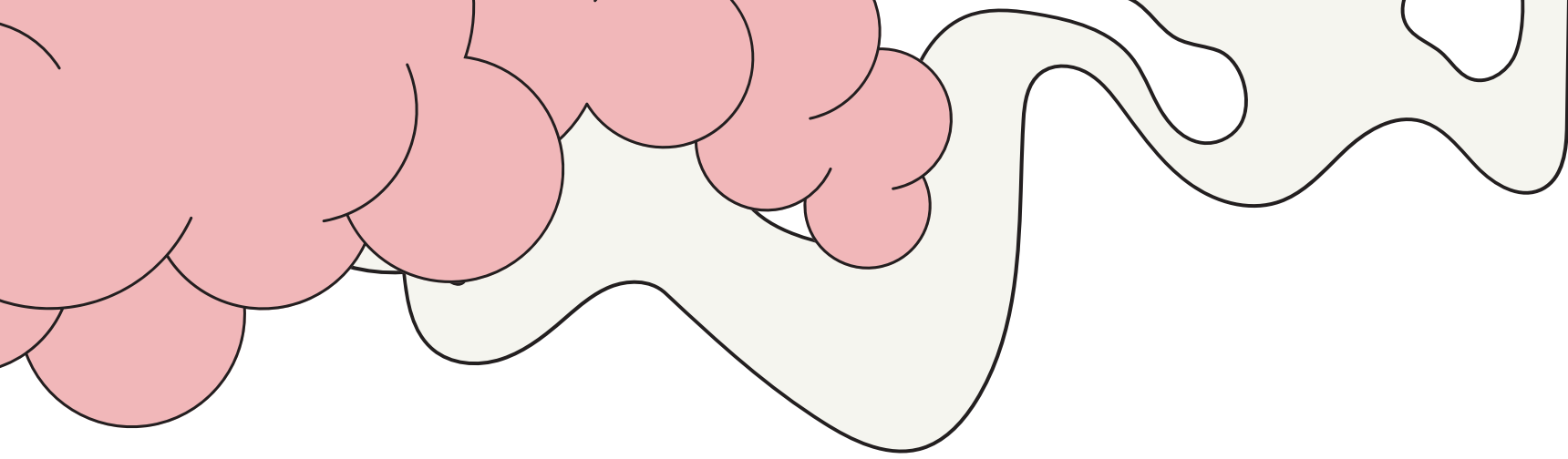


Why It Remains Unsolved?

The vaginal route is a **narrow**, conical channel that forces standard rigid tools into parallel alignment, causing constant **tool clashing** and **surgeon fatigue**.

Current robots fail in this space: multi-port systems cause abdominal wall trauma, while single-port rigid platforms require **straight clearance** that causes severe collisions in the pelvic cavity.





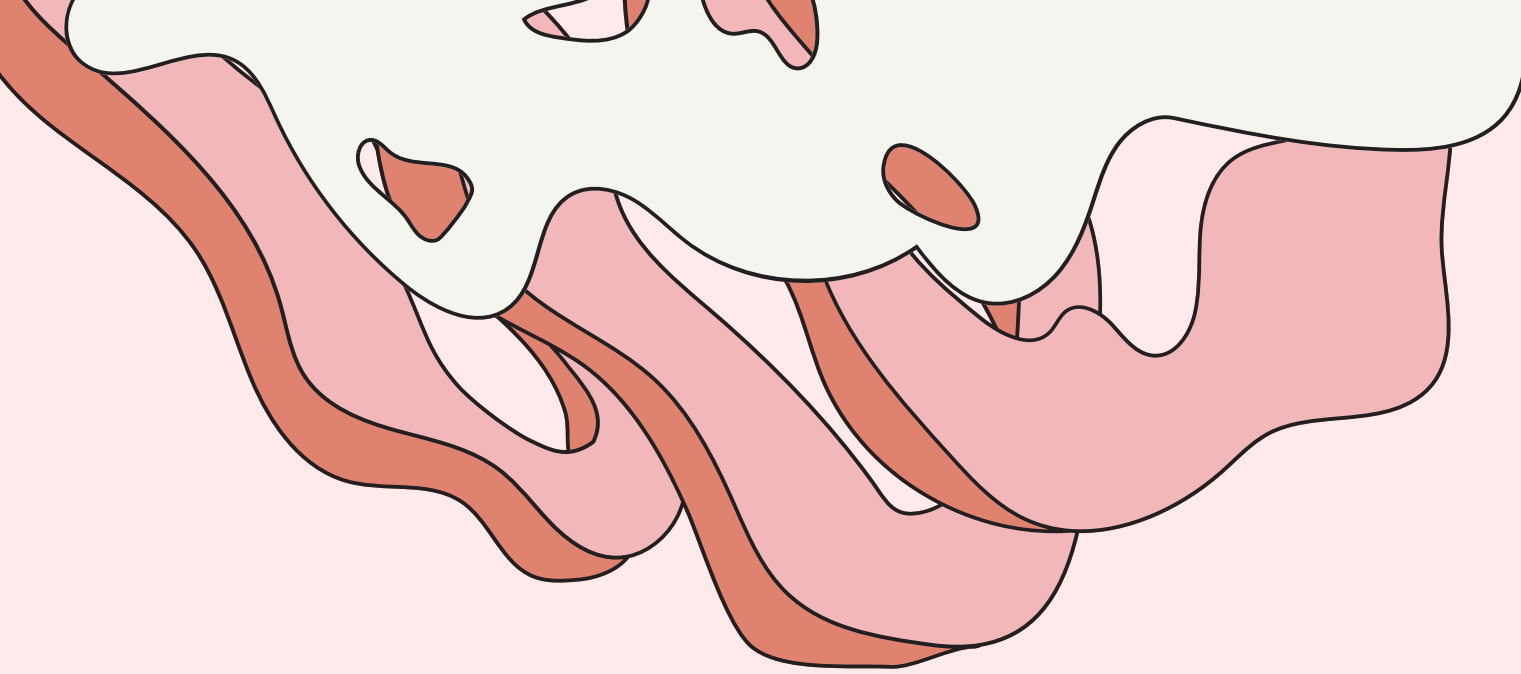
Teleoperated Control (Level 1 Autonomy)

Proposed Robotic Solution

Single-Port Continuum Platform



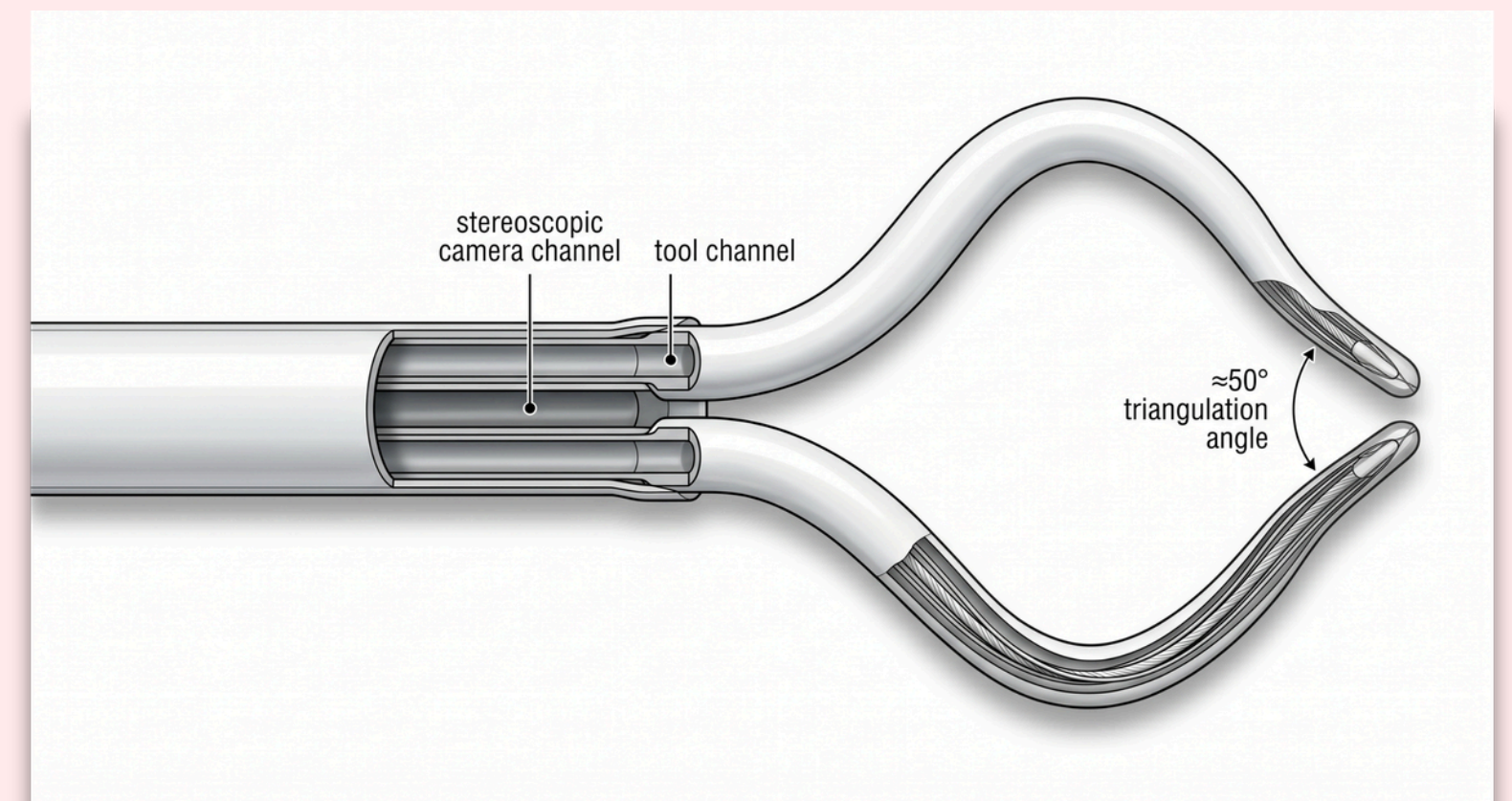
Active Triangulation & Safety

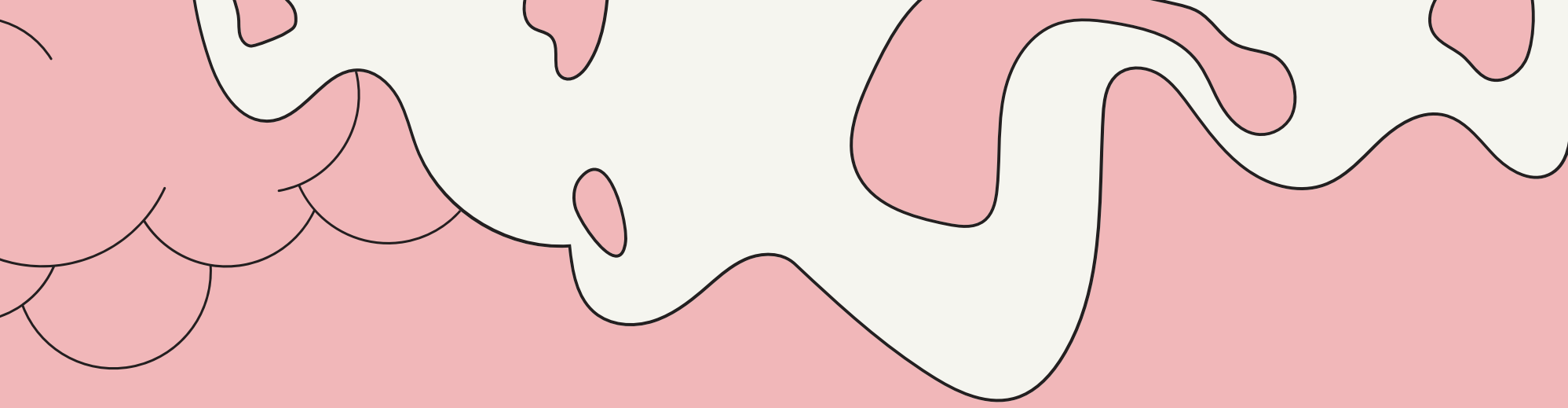


An 18 mm flexible overtube delivering two tendon-driven continuum arms and a 3D stereoscopic camera.

Continuum arms expand outward S-shape to recover a 50 degree internal triangulation angle

Integrated FBG optical sensors detect abnormal tissue contact to prevent perforation.





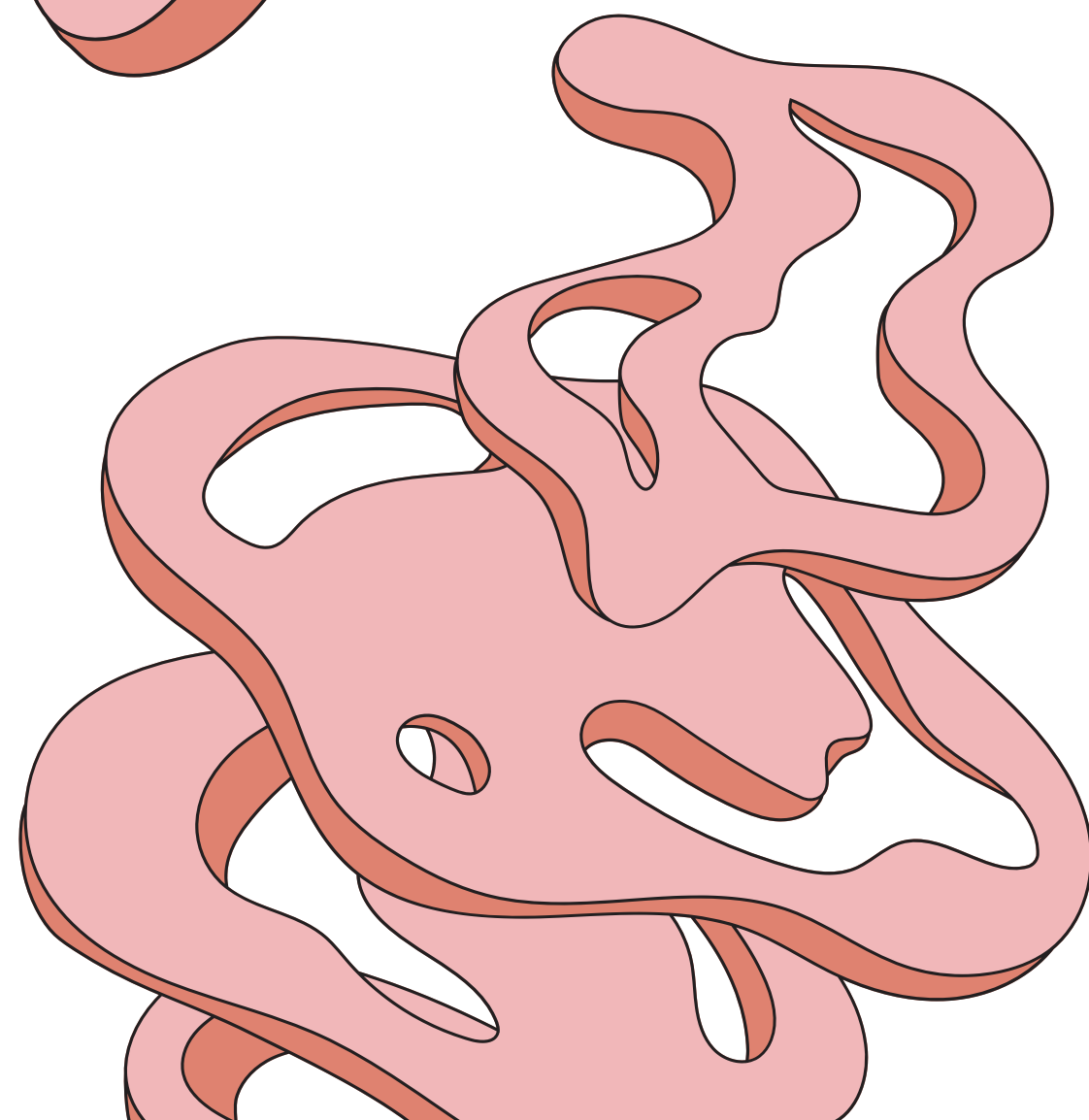
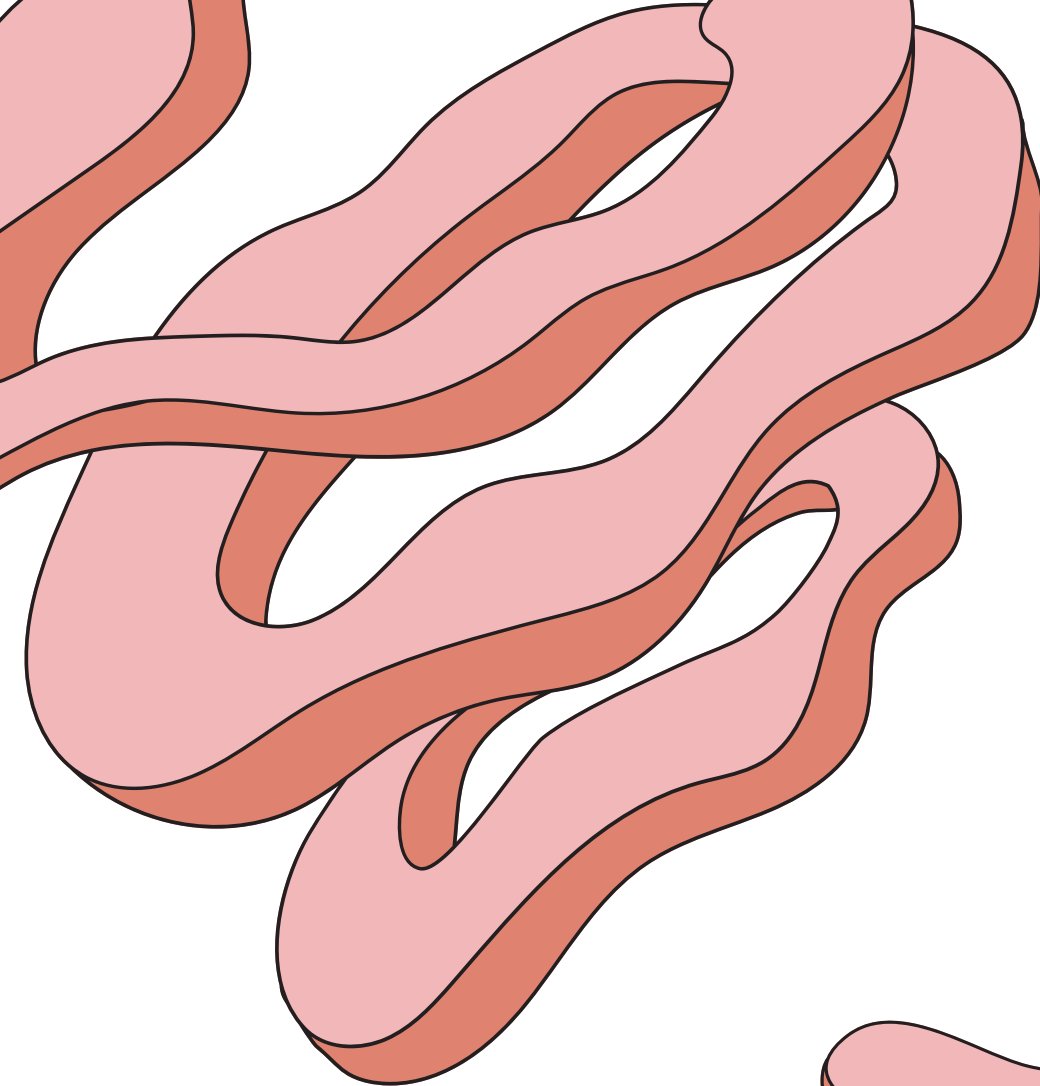
Now the challenges...

Force Interaction Control

- High-load tissue traction causes instrument deflection and position errors.

Sterile Transmission Interface

- Need to transfer high torque from non-sterile motors to sterile disposable arms with zero backlash.



Thank you!

